Stats 102B Homework 1

Christian Jordan

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(1) Angle Between Two Vectors

(a) get_angle() Counterclockwise angle between two vectors

```
get_angle = function(x, y) {
  # Calculates the (counterclockwise) angle between two 2-dimensional vectors

  # Calculate angle
  radians <- acos(dot(x,y) / (Norm(x) * Norm(y)))
  degrees <- radians * (180/pi)

  # Make angle counterclockwise
  cross_prod <- x[1]*y[2] - x[2]*y[1]

  if(cross_prod < 0) {
    degrees <- 360 - degrees
  }

  degrees
}
```

(b) Vector pairs giving 17, 110, and 270 degrees

```
get_angle(c(1,0), c(cos(17 * (pi/180)),sin(17 * (pi/180))))
```

```
## [1] 17
```

```
get_angle(c(1,0), c(cos(110 * (pi/180)),sin(110 * (pi/180))))
```

```
## [1] 110
```

```
get_angle(c(1,0), c(cos(270 * (pi/180)),sin(270 * (pi/180))))
```

```
## [1] 270
```

(2) Matrix Inverse Properties

Given that A^{-1} and B^{-1} exist:

(a) Show that $(A^T)^{-1} = (A^{-1})^T$

$$AA^{-1} = I$$
$$(AA^{-1})^T = I^T$$
$$A^T(A^{-1})^T = I$$
$$(A^T)^{-1}A^T(A^{-1})^T = (A^T)^{-1}I$$
$$(A^{-1})^T = (A^T)^{-1}$$

(b) Show that $(AB)^{-1} = B^{-1}A^{-1}$

$$I = AA^{-1}$$
$$I = AIA^{-1}$$
$$I = A(BB^{-1})A^{-1}$$
$$I = (AB)(B^{-1}A^{-1})$$
$$(AB)(AB)^{-1} = (AB)(B^{-1}A^{-1})$$
$$(AB)^{-1}(AB)(AB)^{-1} = (AB)^{-1}(AB)(B^{-1}A^{-1})$$
$$(AB)^{-1} = B^{-1}A^{-1}$$

(3) Eigenvalue Property

A is kxk matrix and P is kxk orthogonal matrix. Show that P^TAP and A have the same eigenvalues.

$$Ax = \lambda x$$

Let $x = Py$ where y is a vector.

For an orthogonal matrix , $P^T = P^{-1}$, so:

$$x = Py$$
$$P^Tx = P^TPy$$
$$P^Tx = y$$

Using these facts, we have:

$$(P^TAP)y = P^TAx = P^T\lambda x = \lambda P^Tx = \lambda y$$

So, $(P^TAP)y = \lambda y$

So, A and P^TAP have the same eigenvalues λ.

(4) Ellipses

Suppose $c^2 = 5x_1^2 - 2\sqrt{3}x_1x_2 + 7x_2^2$ is the distance from $x = [x_1x_2]^T$ to the origin for $x \neq 0$. Given $c^2 = 1$ and $c^2 = \pi$, find the directions and lengths of the major and minor axes of the ellipses. Sketch the ellipses and discuss what happens if c^2 increases.

Finding the eigenvalues:

$$5x_1^2 - 2\sqrt{3}x_1x_2 + 7x_2^2$$
$$A = \begin{bmatrix} 5 & -\sqrt{3} \\ -\sqrt{3} & 7 \end{bmatrix}$$
$$A = det \begin{bmatrix} 5-\lambda & -\sqrt{3} \\ -\sqrt{3} & 7-\lambda \end{bmatrix} = (5-\lambda)(7-\lambda) - (-\sqrt{3})(-\sqrt{3}) = \lambda^2 - 12\lambda + 32$$
$$\lambda^2 - 12\lambda + 32 = (\lambda - 4)(\lambda - 8)$$
$$\text{So, } \lambda = 4, 8$$

Finding the eigenvectors:

For $\lambda = 4$

$$\begin{bmatrix} 5-4 & -\sqrt{3} \\ -\sqrt{3} & 7-4 \end{bmatrix} = \begin{bmatrix} 1 & -\sqrt{3} \\ -\sqrt{3} & 3 \end{bmatrix} = 0$$
$$x_1 - \sqrt{3}x_2 = 0 \to x_1 = \sqrt{3}x_2 \to x = \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix} \text{ so, } e_1 = \frac{1}{2}\begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}$$

For $\lambda = 8$

$$\begin{bmatrix} 5-8 & -\sqrt{3} \\ -\sqrt{3} & 7-8 \end{bmatrix} = \begin{bmatrix} -3 & -\sqrt{3} \\ -\sqrt{3} & -1 \end{bmatrix} = 0$$
$$-3x_1 - \sqrt{3}x_2 = 0 \to 3x_1 = -\sqrt{3}x_2 \to x = \begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix} \text{ so, } e_2 = \frac{1}{2}\begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$$

In the form of an ellipse equation:

$$c^2 = \lambda_1 y_1^2 + \lambda_2 y_2^2$$
$$1 = \frac{\lambda_1 y_1^2}{c^2} + \frac{\lambda_2 y_2^2}{c^2}$$
$$1 = \frac{y_1^2}{c^2/\lambda_1} + \frac{y_2^2}{c^2/\lambda_2}$$

The lengths of the axes are $\sqrt{c^2/\lambda_1}$ and $\sqrt{c^2/\lambda_2}$

So, for $c^2 = 1$ we have:

The major axis of length $\sqrt{1/4}$ in the direction $\frac{1}{2}\begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}$

The minor axis of length $\sqrt{1/8}$ in the direction $\frac{1}{2}\begin{bmatrix} 1 \\ -\sqrt{3} \end{bmatrix}$

And for $c^2 = \pi$ we have:

The major axis of length $\sqrt{\pi/4}$ in the direction $\frac{1}{2}\begin{bmatrix}\sqrt{3}\\1\end{bmatrix}$

The minor axis of length $\sqrt{\pi/8}$ in the direction $\frac{1}{2}\begin{bmatrix}1\\-\sqrt{3}\end{bmatrix}$

These ellipses are as follows:

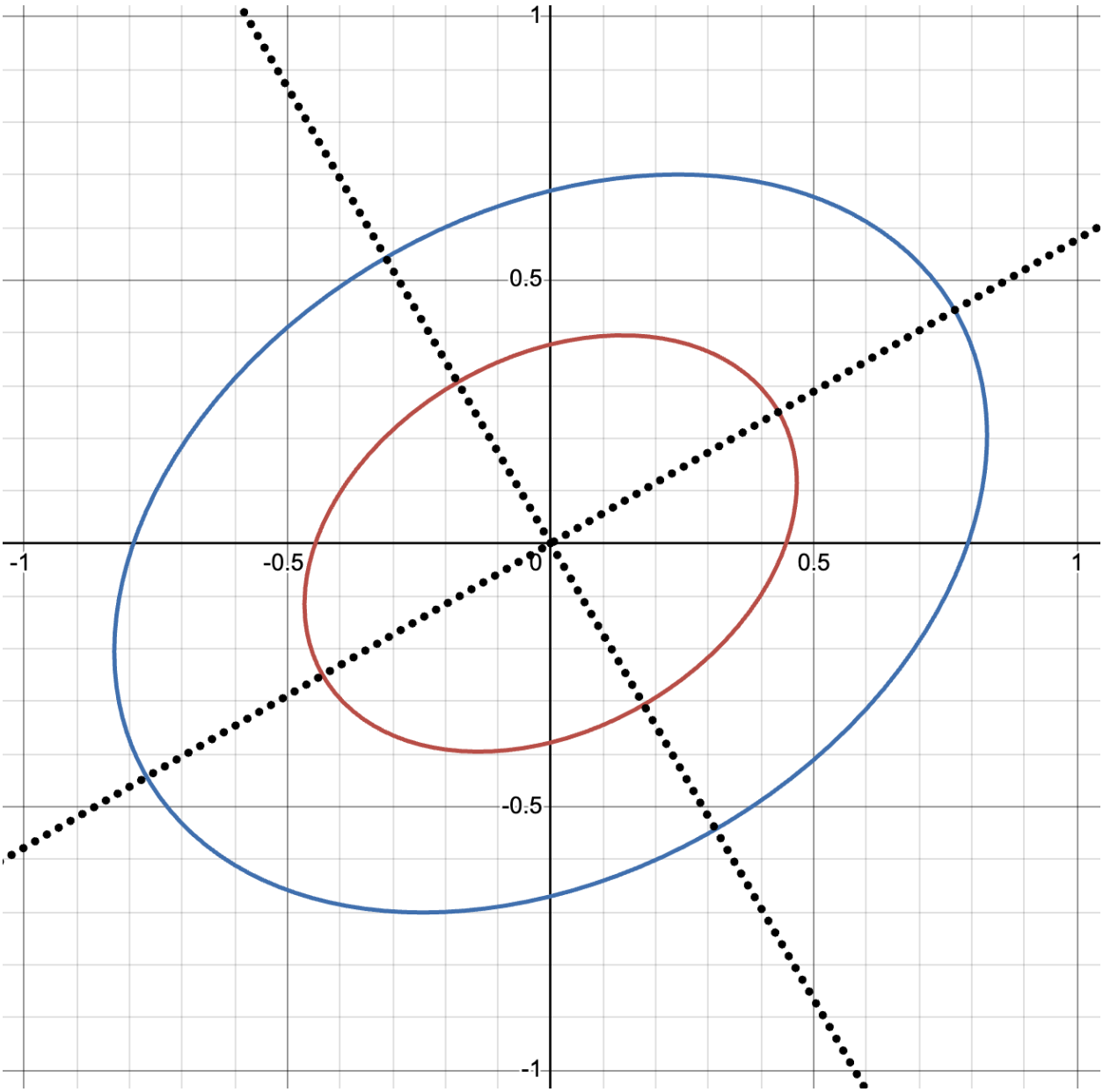


Figure 1: Ellipses. Red is c^2=1 and blue is c^2=pi

The ellipse size increases by a factor of $\sqrt{c^2}$ as c^2 increases

(5) Orthogonal Projection

The data is given below:

```
X = matrix(c(3, -1, 2, 0, 2, 1/2, 1, -3, 1, -1, 2, 1, -3, 4, 1, 2, 1, 1/4, -1, -3, 5, 2, 7, 3/2),
           nrow = 6, ncol = 4)

y = matrix(c(-2, -7, -1, 3, 2, 1),
           nrow = 6, ncol = 1)
```

(a) Find orthogonal projection $\pi_X(y)$ of y onto X.

Use $\pi_X(y) = X(X^T X)^{-1} X^T y$

```
# Projection of y onto X
proj_yX <- X %*% solve(t(X) %*% X) %*% t(X) %*% y
proj_yX
```

```
##            [,1]
## [1,] -2.5743490
## [2,] -6.6383683
## [3,]  0.5358446
## [4,]  0.8940434
## [5,]  2.0042680
## [6,] -0.9910930
```

(b) What is the projection matrix and projection error?

```
# Projection matrix
X %*% solve(t(X) %*% X) %*% t(X)
```

```
##              [,1]         [,2]       [,3]        [,4]        [,5]        [,6]
## [1,]  0.958588483  0.002381394 0.14746088 -0.09853405 -0.06958619 -0.05826684
## [2,]  0.002381394  0.966670790 0.04296819  0.08034886 -0.09391559  0.12284283
## [3,]  0.147460877  0.042968186 0.39516711  0.23510855  0.39955877  0.02226758
## [4,] -0.098534051  0.080348859 0.23510855  0.59751345  0.05474114 -0.40749675
## [5,] -0.069586194 -0.093915589 0.39955877  0.05474114  0.59421455  0.25459269
## [6,] -0.058266840  0.122842828 0.02226758 -0.40749675  0.25459269  0.48784561
```

```
# Projection error
error <- y - proj_yX
error
```

```
##              [,1]
## [1,]  0.574348982
## [2,] -0.361631719
## [3,] -1.535844622
## [4,]  2.105956578
## [5,] -0.004267953
## [6,]  1.991092967
```

(c) Distance between y and $\pi_X(y)$.

```r
# Distance from y to projection
sqrt(sum(error^2))
```

```
## [1] 3.349477
```

(6) Linear Model

```
data <- read.csv("data1.csv")
```

(a) Find the least squares estimator $\hat{w}$ for the linear model.

Use $\hat{w} = (X^T X)^{-1} X^T y$

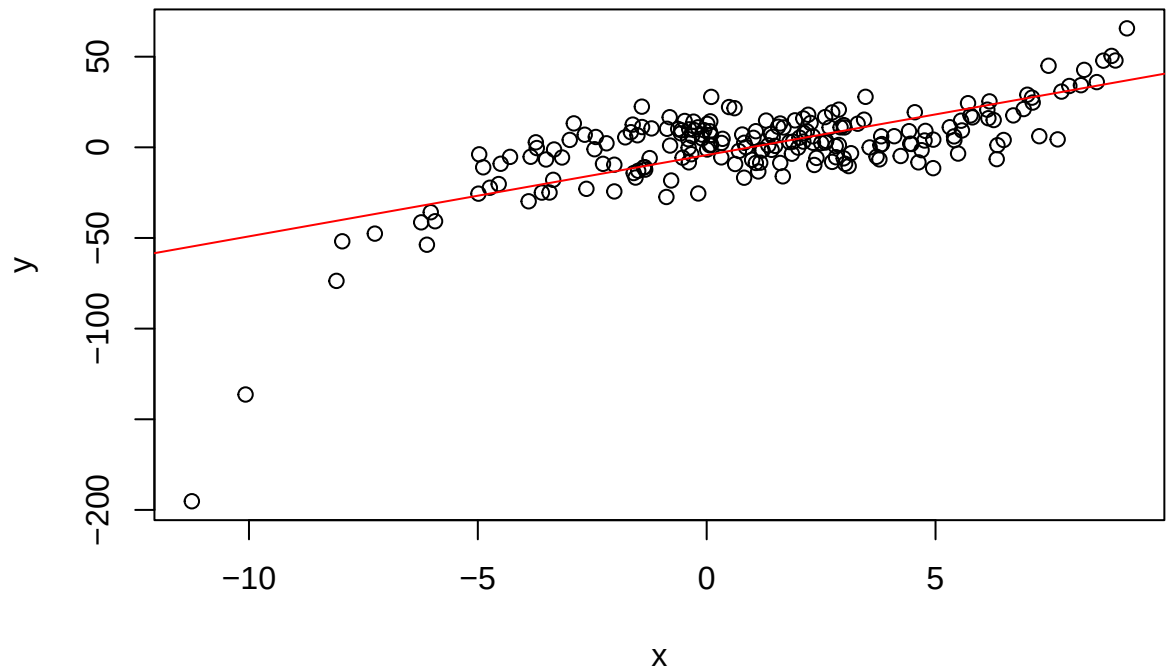
```
x <- data$x
X <- cbind(1, x)
y <- data$y

w <- solve(t(X) %*% X) %*% t(X) %*% y
w
```

```
##        [,1]
##   -4.269400
## x  4.485172
```

The intercept is -4.269 and the beta1 coefficient is 4.485

(b) Plot the model. Does it fit the data well?

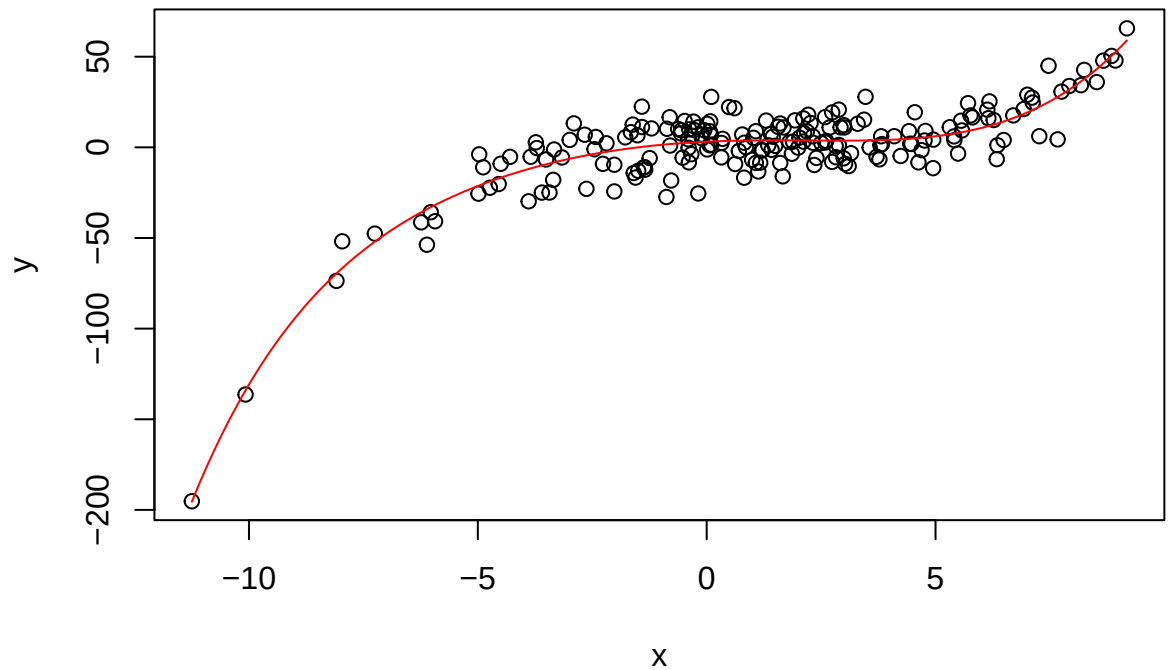
```
plot(x, y)
abline(a = w[1], b = w[2], col = "red")
```



The model does not fit the data well. We see that the data has some curvature to it, but we are trying to use a linear model to model it. It is an especially poor fit at the extremes of the data.

(c) Try using a 6th order polynomial fit.

```
model <- lm(y ~ poly(x, 6, raw = TRUE))
plot(x, y)
x_grid <- seq(min(x), max(x), length.out = 200)
y_pred <- predict(model, newdata = data.frame(x = x_grid))

lines(x_grid, y_pred, col = "red")
```



(d) Does the above model fit the data well? Use LOOCV to determine the optimal k-th order polynomial model that best fits the data.

The 6th order polynimal model is a much better fit for the data than the linear model. It fits well.

```
mse <- numeric(10)
n <- nrow(data)

for (k in 1:10) {
  errors <- numeric(n)

  for (i in 1:n) {
    train <- data[-i, ]
    test  <- data[i, ]

    model <- lm(y ~ poly(x, k, raw = TRUE), data = train)

    pred <- predict(model, newdata = test)

    errors[i] <- (test$y - pred)^2
  }

  mse[k] <- mean(errors)
}
```

```r
rbind(mse)
```

```
##         [,1]     [,2]     [,3]     [,4]     [,5]     [,6]     [,7]     [,8]
## mse 338.5839 292.6752 113.6875 117.4611 111.6157 112.5274 115.6835 127.8693
##         [,9]    [,10]
## mse 474.0375 687.2479
```

From the LOOCV, we see that the 5th order polynomial is the one with the smallest MSE and is the best fit for the data.